

Student ID:

Lab 03 Task: The MAGA Headquarters

Your company, Make Abdullahpur Great Again (MAGA), is expanding its headquarters and requires a new network design to support various departments. The MAGA headquarters consists of the following departments across multiple floors: **Engineering** (Floor 1), **Marketing** (Floor 2), **Sales** (Floor 3), **Administration** (Floor 4), **Management** (Floor 5).

Suppose they all share the prefix **192.(i).(i+3) % 10**. Below are the required devices for each of the departments:

$E(n) = 33$	(Engineering)
$M(n) = 7$	(Marketing)
$S(n) = 14$	(Sales)
$A(n) = 12$	(Administration)
$G(n) = 30$	(Management)

where, i is the last two digits of your id

Task Requirements

1. **Topology Implementation:** Implement and configure the topology. For each network, you need to add a minimum of 3 end devices.
2. **DHCP Configuration:**
 - Devices should get their IP automatically from the DHCP server.
 - **Engineering, Marketing, and Sales:** Devices should use a separate dedicated DHCP server for automatic IP assignment.
 - **Admin and Management:** Devices should use a Router as a DHCP server.
3. **NAT Configuration:**
 - Your company is connected to the internet via an "Internet Router".
 - You have purchased only **7 Global IPs**.
 - The Global IP address range starts from **110.i.(i+1).0/24**.
 - **Your own PC** must have a specific local IP that maps to its own exclusive Global IP
 - **All other devices** must share the remaining six Global IP addresses for accessing the internet

Lab 03 Grading Criteria

1. **Topology & Router Configuration (20 pts)**
 - Correct topology design (5 pts)
 - Correct router interfaces and basic setup (15 pts)
2. **DHCP Configuration (20 pts)**
 - Correct separate DHCP server configuration (10 pts)
 - Correct Router-as-DHCP configuration (10 pts)
3. **NAT Configuration (20 pts)**
 - Correct NAT Configuration (10 * 2 pts)
4. **Routing (10 pts)**
 - Correct Routing configuration to ensure connectivity (10 pts)
5. **Total: 70 pts**